

Question-Answer

Q1. What type of cable is used to connect the Ethernet interface on a host PC to the Ethernet interface on a switch?

Answer: Straight-through cable

Q2. What type of cable is used to connect the Ethernet interface on a switch to the Ethernet interface on a router?

Answer: Straight-through cable

Q3. What type of cable is used to connect the Ethernet interface on a router to the Ethernet interface on a host PC?

Answer: Crossover cable

Q4. What would happen if you answered yes to the question, "System configuration has been modified. Save?"

Answer: If the answer is "yes," the router saves the current (possibly incomplete or incorrect) running configuration to NVRAM as the startup configuration. This means the configuration will reload with those saved settings instead of starting clean.

Q5. Why would you want to disable DNS lookup in a lab environment?

Answer: To prevent the router from trying to translate mistyped commands into domain names, which causes long delays.

Q6. What would happen if you disabled DNS lookup in a production environment?

Answer: Users would be unable to use hostnames in commands and would need to use IP addresses instead. However, this typically wouldn't break operations since routers primarily use IP addresses.

Q7. Why is it not necessary to use the enable password *password* command?

Answer: Because enable secret provides stronger, encrypted password protection, making enable password redundant.

Q8. When does this banner (banner motd) display?

Answer: The banner displays before login, whenever anyone connects to the router via console, Telnet, or SSH.

Q9. Why should every router have a message-of-the-day banner?

Answer: To display legal warnings and unauthorized access notices, helping deter unauthorized users and provide legal protection.

Q10. What is a shorter version of this(copy running-config startup-config) command?

Answer: copy run start

Q11. From the host attached to R1, is it possible to ping the default gateway?

Answer: Yes

Q12. From the host attached to R2, is it possible to ping the default gateway?

Answer: Yes

Q13. From the router R1, is it possible to ping R2 using the command ping 192.168.2.2?

Answer: Yes

Q14. From the router R2, is it possible to ping R1 using the command ping 192.168.2.1?

Answer: Yes

Q15. What is missing from the network that is preventing communication between these devices?

Answer: The routers are not configured with static or dynamic routing, so they don't know how to reach remote networks beyond their directly connected interfaces. A routing protocol or static routes is needed for end-to-end connectivity between hosts on different networks.

Screenshots

```
R1#show running-config
Building configuration...
```

```
Current configuration : 857 bytes
!
version 12.3
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCil
!
!
!
!
!
no ip cef
no ipv6 cef
```

```
R1#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
```

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.1.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	NVRAM	administratively down	down
Serial0/0/0	192.168.2.1	YES	manual	up	up
Serial0/0/1	unassigned	YES	NVRAM	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
R2#show running-config
Building configuration...
```

```
Current configuration : 829 bytes
!
version 12.3
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R2
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCil
!
!
!
!
!
ip cef
no ipv6 cef
```

```
R2#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.2.0/24 is directly connected, Serial0/0/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
```

```
R2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.3.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0/0	192.168.2.2	YES	manual	up	up
Serial0/0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down